

Silent Failure Modes in DFT Screening of Oxygen-Evolution Electrocatalysts

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Problem

Earth-abundant oxide electrocatalysts for the oxygen evolution reaction (OER) are routinely screened by DFT, but the screens are rarely stress-tested against their own methodological choices. This project builds a multi-fidelity screen — a machine-learned interatomic potential (Meta's UMA) proposes high-entropy-alloy oxide candidates, and Quantum ESPRESSO (PBE+U) re-ranks the top candidates on rutile(110) surfaces with OH/O^*OOH intermediates and computational-hydrogen-electrode referencing — and then turns the same rigor on the screen itself. The result is a project about what silently breaks a DFT overpotential ranking, not just what the ranking says.

What broke, and what it means

Before quoting results, a set of robustness checks was pre-registered (i.e., the acceptance/failure criteria were written down before the calculations ran, so the outcome couldn't be reinterpreted after the fact). Two of them fired:

- **U-sensitivity (project-falsifying).** The initial headline — that an earth-abundant rutile (Cr) outperforms the noble-metal anchors RuO_2/IrO_2 — depended on a Hubbard-U value the project had not derived from anything. Re-running the same ranking across a defensible U range moved $\eta(Cr)$ by **1.12 V**, roughly $7.5\times$ the pre-registered falsification threshold. The claim was withdrawn, on schedule and in writing, rather than softened.
- **Magnetic multistability.** Several adsorbate relaxations that reported clean BFGS convergence had actually settled into a metastable magnetic solution — a basin the optimizer never left. One case sat 175–405 meV above the true minimum despite a textbook-looking force trace. A systematic audit found this in 5 of 7 magnetic 3d endmembers checked so far; it does not affect the two non-magnetic noble-metal anchors.
- **A symmetry-constrained relaxation trap,** found by comparing production (mirror-plane-constrained) adsorbate geometries against orientationally unconstrained restarts: one anchor's *OOH intermediate was ~ 0.29 eV lower off the mirror plane, moving that anchor's overpotential from outside the literature range to inside it. The effect turned out to be coverage-conditional, not tier-wide, once tested at the correct unit cell — itself a finding that needed disconfirming, not assuming.

None of this was found by getting a “wrong” number and going looking for a bug — every one of these was a **clean-looking result** (converged forces, sensible geometry, plausible energy) that turned out to be silently wrong for a reason the standard convergence checks don't see. That is the actual subject of the STS entry: cataloguing and, where possible, generalizing the failure modes, not just reporting a corrected overpotential table.

Status now

The DFT production cell and coverage choice are now settled by a 22-point internal consistency gate (all SCFs agree; no undetected basin drift). An ACCESS Explore allocation (CHE260157, 100,000 Anvil CPU-SUs, PI) is provisioning to run the next campaign — a crossed coverage $\times$ symmetry $\times$ basin-restart study across an 8-oxide tier — launching in the next two weeks. In parallel, melt access is confirmed at Fort Wayne Metals (arc-melting, six to eight high-entropy-alloy-oxide candidates) with an electrochemistry bench at Purdue for OER benchmarking against the NiFe-LDH standard, so the computational rankings will face a real measurement before the project's mid- October data freeze.

What I'm asking for

This is a Regeneron Science Talent Search entry (due November 5, 2026); any input from a mentor is purely advisory — all report prose stays the student's own, per STS policy. I'm looking for a brief conversation and, if the project seems sound, occasional feedback through the fall: specifically, whether the failure modes above are recognized patterns in the field, whether the diagnostic protocol is sound, and where it's still missing something.

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